

Derrick Quinn

Curriculum Vitae

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in derrick-quinn

Overview

Area of interest: Algorithm-hardware-system co-design for Neural Information Retrieval systems.

Vision: My research philosophy is driven by the belief that the interactions between algorithms, hardware, and systems must guide the design of next-generation computer systems. Following this approach, my vision is to develop generalized and scalable full-stack architectures for neural information retrieval systems, expanding their capability, efficiency, and scalability across diverse computing environments.

Education

- 2024–2028 **Ph.D. Electrical and Computer Engineering**, *Cornell University*, Ithaca, NY
Advisor: Mohammad Alian
- 2018–2024 **B.S. Computer Science**, *University of Kansas*, Lawrence, KS
- 2018–2024 **B.S. Mathematics**, *University of Kansas*, Lawrence, KS

Professional Experience

- 2024–Current **Research Assistant**, *Department of Electrical and Computer Engineering, Cornell University*, Ithaca, NY
- 2022–2024 **Research Assistant**, *Department of Electrical Engineering and Computer Science, University of Kansas*, Lawrence, KS
- 2021–2022 **Algebra Tutor and Class Assistant**, *Kansas Algebra Program; Department of Mathematics, University of Kansas*, Lawrence, KS
- 2020 **Engineering Intern**, *Dell Technologies*, Overland Park, KS

Publications

- [5] **LongSight: Compute-Enabled Memory to Accelerate Large-Context LLMs via Sparse Attention**, *MICRO: International Symposium on Microarchitecture (2025)*, Seoul, KR
Derrick Quinn, Emine Ezgi Yücel, Jinkwon Kim, José F. Martínez, Mohammad Alian
- [4] **DReX: Accurate and Scalable Dense Retrieval Acceleration via Algorithmic-Hardware Codesign**, *ISCA: International Symposium on Computer Architecture (2025)*, Tokyo, JP
Derrick Quinn*, Emine Ezgi Yücel*, Martin Prammer, Zhenxing Fan, Kevin Skadron, Jignesh Patel, José F. Martínez, Mohammad Alian
- [3] **Compute-Enabled CXL Memory Expansion for Efficient Retrieval Augmented Generation**, *IEEE Micro*, Magazine
Derrick Quinn, Neel Patel, Mohammad Alian
- [2] **Accelerated Retrieval Augmented Generation**, *ASPLOS: International Conference on Architectural Support for Programming Languages and Operating Systems (2025)*, Rotterdam, NL
Derrick Quinn, Mohammad Nouri, Neel Patel, Alireza Salemi, Sukhan Lee, Hamed Zamani, Mohammad Alian
- [1] **XFM: Accelerated Software-Defined Far Memory**, *MICRO: International Symposium on Microarchitecture (2023)*, Toronto, Ontario, CA
Neel Patel, Amin Mamandipoor, Derrick Quinn, Mohammad Alian